

A conjecture for the refinement of the Aldous-Shepp theorem

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This document is intended to be the necessary background for a demo of the package `linearframe-work.py`.

This work was done in the laboratory of Prof. Jeremy Gunawardena in the department of systems biology at Harvard Medical School, as the summer project of Malachy Guzmann, an undergraduate at Carleton College, supervised by Emre Alca.

1 Introduction

I believe it is the responsibility of scientists not to only collect data, but to interpret it. The goal is to find real patterns within the data—to perform meaningful compression. These real patterns are about relationships. I want to understand what relationships drive a result and how said result can be explained. In short, I am interested in mechanisms.

Modern biology has been driven by the development of incredible instruments providing unimaginable amounts of data. As the use of mass spectrometers, DNA sequencers, and electron microscopes has become commonplace, new experimental results have long since outpaced theoretical developments—leaving much of this data uninterpreted or interpreted only as an afterthought [7].

In my home discipline of physics, there has been a long tradition of collaboration between theorists and experimentalists. A theorist proposes a theory and an experimentalist collaborates with them to test that theory. This is theory guiding experiments—the experiment is informed by the theory it intends to test. Often the theory does not quite agree with the experimental results and the conversation continues. This is the process of theory motivating experiment and experiment informing theory or forward modelling [3].

The incredible measurement devices mentioned above, and their relative ease of use, has created a glut of data that is desperately in need of interpretation. When experiment follows theory, it is usually very clear whether a result supports or falsifies a theoretical claim. The inverse problem is much more opaque. Often the best tools we have when using inverse modelling are statistical correlations. This kind of ‘reverse modelling’ has been useful in modern biology to prove the existence of new molecules or objects or interactions, but has failed to find the kind of principles that traditional forward modelling methods have in the past, such as chemotaxis or homeostasis [3].

In this note, I propose a generalization of one quantity which we can calculate from experimental data and use to constrain our choice of model: the Aldous-Shepp theorem [1]. The Aldous-Shepp theorem places a lower bound on the number of states a stochastic system (specifically, a Markov system) must have to generate a given dwell-time distribution (a distribution of waiting-times between actions) using it’s randomness parameter: the variance over the mean squared. Here, I propose a generalization of Aldous-Shepp theorem to a similar statement of higher moments of a dwell-time distribution. Proving this conjecture here, only build up the conjecture itself, and explore it numerically in a python notebook.

2 Background

2.1 Understanding the experiment: dwell-time distributions of molecular motors.

This project follows the work of Moffitt and Bustamante [4]. Moffitt and Bustamante explore a number of experiments which measure dwell-time distributions (distributions of waiting times between measurable actions) of single-molecule systems involving enzyme kinetics (bio-molecules whose chemical composition does not change during a reaction). In this note, we will focus on a single experimental example, the molecular motor, although the results generalize to any dwell-time distribution. Molecular motors are little walkers made up of a single molecule (the enzyme) that move various kinds of cargo along structures called microtubules within cells. We will not worry about microtubules in this note—though they are very interesting. For now, consider a molecular motor to be like a walking wind-up toy.

The molecular motor is a very slow wind up toy—it’s proverbial spring may need to be wound many times before it can take a single step (see figure 1A for a cartoon representation). But this spring does not get wound deterministically—it can spontaneously and at random wind and unwind—the molecular motor binds or unbinds with ‘fuel molecules’ called adenosine triphosphate or ATP at random. In taking each step, the motor ‘consumes’ the fuel molecules bound to it (by converting ATP into adenosine diphosphate or ADP), and must bind to new fuel molecules in order to take the next step. The toy model in figure 1 requires that two fuel molecules be bound to the motor in order to take a step, but the mechanism can be much more complicated in reality. Our goal is to learn more about the structure of this mechanism using experimental data.

What is measurable with these walkers is when they take steps—or rather the time between steps. Measuring this time between steps make up our samples of the motor’s dwell-time distribution. The dwell-time distribution is a random, inheriting it’s stochasticity from the stochasticity of the chemical reactions themselves. We do not have a good way of measuring the internal dynamics of the walker—i.e. how many ‘fuel molecules’ need to be bound, or if any other intermediate reactions are necessary for each step. We are explicitly interested in what we can learn about the structure of the walker’s internal dynamics from its dwell-time distribution.

2.2 The Aldous-Shepp theorem

One focus of the Gunawardena group is modelling chemical reaction networks as Markov chains—and equivalently as weighted directed graphs—to make quantitative claims about their mechanistic behaviour [2]. We can think of the internal dynamics of a walker as a traversal through a continuous-time Markov chain. Markov chains are stochastic processes with no memory—a Markov system’s next step depends only on it’s current state and not on its complete history. In order to take a step, the motor’s internal dynamics must traverse through some Markov chain from some source state i to some other target state q . We call the time required to traverse the Markov chain (or equivalently, the graph) from i to q is called a first-passage-time (FPT). The FPT of a continuous time Markov system is a random variable itself since continuous time Markov systems are random in both *where* and *when* they transition between states. Each FPT can be considered a single dwell-time, and dwell-time distribution is the FPT distribution. Recently, the Gunawardena group has found explicit formulas for the moments of the FPT distribution of any weighted-directed graph [5]. For more details about those formulas, please do see [6] and [5].

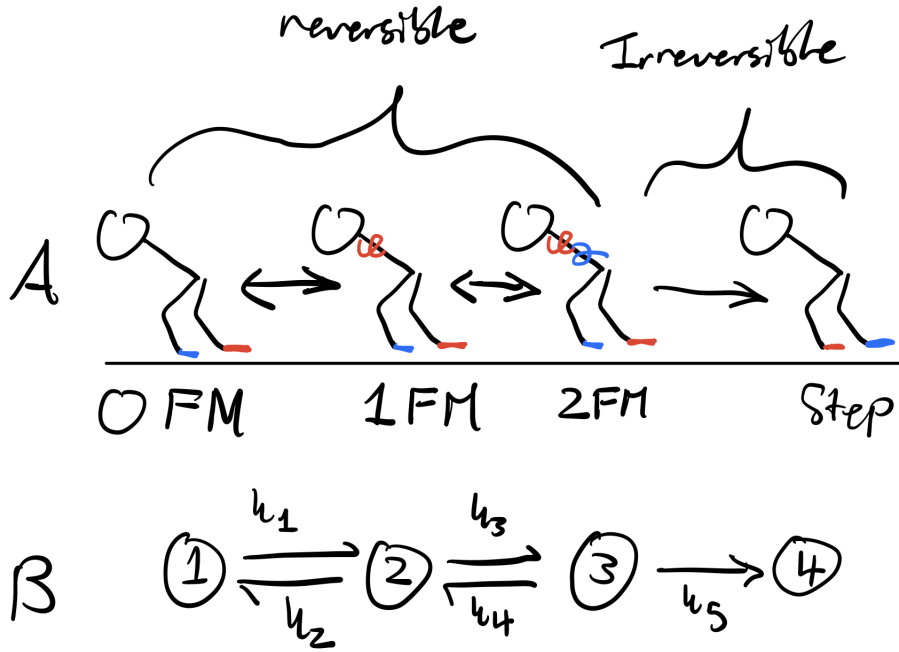


Figure 1: A: Consider a toy model of the molecular motor as if we had perfect access to it's mechanism. In this toy model, the motor only takes a step when two fuel molecules are bound to it—as if a wind-up toy needs to have two rotations of its crankshaft for each step. Each molecular transition (binding and unbinding) is a reversible process which happens randomly with some rate (number of times each second). Each step the motor takes is a non-reversible transition, 'consuming' the fuel molecules in the process. In order to take the next step, the motor must have two new fuel molecules bind to it.

B: A graph theoretic model representing the walker or it's associated continuous-time Markov Chain. We consider state 1 to be the walker with no fuel molecules bound to it, 2 the state with one fuel molecule bound, 3 the state with 2 fuel molecules bound, and 4 representing the motor just after having taken a step. This graph is both weighted and directed. Each transition has a certain rate k_i (units of inverse seconds, s^{-1}).

In reality, we do not know what the graph looks like. Our goal is to find quantitative constraints on the choice of graph given a collection of samples of the dwell-time distribution.

Moffitt and Bustamante also model the internal dynamics of the walker as a Markov chain, but not as explicitly as a graph. They make use of a result in the theory of stochastic processes: the Aldous-Shepp theorem [1]. Aldous and Shepp were interested in using the dispersion of the dwell-time distribution to limit the complexity of model choice. Ideally, such a measure would be a dimensionless and normalized measure of dispersion. They chose to use 'randomness parameter' or the ratio of the variance in the FPT distribution over its mean squared. The randomness parameter can also be thought of as the squared coefficient of variation for the FPT distribution. The Aldous-Shepp theorem states that the 'randomness parameter'—of a Markov process with n states is bounded below by $1/(n+1)$. Let $t(G)_{i,q}$ be the random variable representing a FPT of our Markov process G on n vertices from some initial state i to a target state q . For our toy walker, there is a loop here—the initial vertex is having just taken a step, and the

target vertex is taking a step. For brevity, and because of the well-defined nature of these source and target vertices, we write our FPT variable as just $t(G)$. Where we write:

$$r \equiv \frac{\langle t(G)^2 \rangle - \langle t(G) \rangle^2}{\langle t(G) \rangle^2} \geq \frac{1}{n-1}. \quad (1)$$

Do note that the randomness parameter is a ratio of two quantities with dimensions of time, making the units of the randomness parameter dimensionless time. The Aldous-Shepp theorem enables us to find an absolute lower bound the number of states in a Markov process with the randomness parameter r .

$$r^{-1} + 1 = n_{\min} \leq n.$$

The Aldous-Shepp theorem is a quantitative constraint on the complexity of our model of the walker's internal dynamics. In fact, the Aldous-Shepp result is often used to estimate the number of states in a graph. I find this concerning—it provides a *lower bound* on the number of states, but not a good estimate of the number of states itself.

3 Analysis: Generalizing the Aldous-Shepp theorem

We come in to this project with three goals. The first goal was to find a graph-theoretic proof of the Aldous-Shepp theorem. The second goal was to test whether a quantity similar to r but using higher moments of the FPT distribution than just the first and the second provides a bound containing additional structural information about the underlying graph. The third goal is to see whether our explicitly graph-theoretic approach is further illuminating from the perspective of mechanism.

The measure we chose to define at the 'generalized randomness parameter' for some graph G is defined as:

$$s_m(G) = \frac{\langle t(G)^m \rangle}{\langle t(G) \rangle^m} \quad (2)$$

where $\langle t(G)^m \rangle$ is the m -th moment of the FPT distribution on G from i to q . We can restate the Aldous-Shepp theorem (equation 1) as

$$s_2 = \frac{\langle t(G)^2 \rangle}{\langle t(G) \rangle^2} \geq \frac{n}{n-1}.$$

We want to find some lower bound on $s_m(G)$ regardless of the particular weights in G .

Our first approach was to look at where the initial Aldous-Shepp bound arises from. Aldous and Shepp showed that the $r = 1/(n+1)$ for a particular Markov process on n vertices called the Erlang process [1]. Aldous and Shepp also showed that this Erlang distribution is the least-variable variable distribution (meaning it has the lowest randomness parameter). The Erlang process on n vertices (E_n) is a line of irreversible transitions, with each transition having the same rate (see figure ??). An undergraduate student I was supervising and I were able to derive an expression for the generalized randomness parameter on an arbitrary Erlang process

$$s_m(E_n) = \frac{\prod_{j=0}^{m-1} (n-1+j)}{(n-1)^m} \quad (3)$$

This proof can be found, as written by Malachy, below. The proof is straightforward, leaning mostly on a simple counting argument. This same result can be found using other known expressions for the moments

of the Erlang distribution. Thus, we pose the conjecture that the generalization of the Aldous-Shepp theorem would be

$$s_m(G_n) \geq s_m(E_n) = \frac{\prod_{j=0}^{m-1} (n-1+j)}{(n-1)^m}. \quad (4)$$

Now we can turn to our python notebook demo and explore this result numerically.

References

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3 Higher Moments and the Erlang Bound

The standard randomness parameter, r , is given by the variance divided by the squared mean of the first passage time. For a process on n vertices,

$$r = \frac{\langle t^2 \rangle - \langle t \rangle^2}{\langle t \rangle^2} = \frac{\langle t^2 \rangle}{\langle t \rangle^2} - 1$$

And Aldous-Shepp tells us that

$$\begin{aligned} \frac{\langle t^2 \rangle}{\langle t \rangle^2} - 1 &\geq \frac{1}{n-1} \\ \frac{\langle t^2 \rangle}{\langle t \rangle^2} &\geq \frac{n}{n-1} \end{aligned}$$

This bound is achieved by the Erlang graph. Let's analogize this to the higher moments:

Definition 18. Let the r_m be the m -th randomness parameter, given by $r_m = \frac{\langle t^m \rangle}{\langle t \rangle^m} - 1$

Definition 19. For easier notation, let's also define $s_m = \frac{\langle t^m \rangle}{\langle t \rangle^m}$, so $s_m = r_m + 1$.

Definition 20. Let E_n denote the Erlang graph on n vertices, with uniform weight w :

$$1 \xrightarrow{w} 2 \xrightarrow{w} \dots \xrightarrow{w} n$$

Definition 21. Another piece of easier notation (by Emre): let

$$f(p, q, r) = \sum_{F \in \Phi_{\{p, q\}}: r \rightsquigarrow p} w(F)$$

That is the sum of the weight of spanning forests doubly rooted at p and q such that there's a path from r to p .

Definition 22. Let $g(p, q, r) = |\Phi_{\{p, q\}} : r \rightsquigarrow p|$. Observe that every spanning 2-forest has $n - 2$ edges and so has weight $w(n - 2)$. So

$$g(p, q, r) = \frac{f(p, q, r)}{w \cdot (n - 2)}$$

Proposition 23. Here is an explicit formula in terms of n and m for the m -th randomness parameter of the Erlang process on n vertices:

$$s_m(E_n) = \frac{(n + m - 2)!}{(n - 2)!(n - 1)^m}$$

Proof. Observe that

$$\begin{aligned} s_m = \frac{\langle t^m \rangle}{\langle t \rangle^m} &= \frac{m! \sum_{(j_1, \dots, j_m)} \left(\prod_{u=1}^m [f(j_u, q, j_{u-1})] \right)}{\left(\sum_{F \in \Phi_{\{q\}}} w(F) \right)^m} \cdot \frac{\left(\sum_{F \in \Phi_{\{q\}}} w(F) \right)^m}{\left(\sum_{j=1}^{n-1} f(j, q, i) \right)^m} \\ &= \frac{m! \sum_{(j_1, \dots, j_m)} \left(\prod_{u=1}^m [f(j_u, q, j_{u-1})] \right)}{\left(\sum_{j=1}^{n-1} f(j, q, i) \right)^m} \\ &= \frac{m! \sum_{(j_1, \dots, j_m)} \left(\prod_{u=1}^m [w(n-1) \cdot g(j_u, q, j_{u-1})] \right)}{\left(\sum_{j=1}^{n-1} w(n-1) \cdot g(j, q, i) \right)^m} \end{aligned}$$

$$= \frac{m! \sum_{(j_1, \dots, j_m)} \left(\prod_{u=1}^m [g(j_u, q, j_{u-1})] \right)}{\left(\sum_{j=1}^{n-1} g(j, q, i) \right)^m}$$

Let's think about the denominator first. $i = 1$ and $q = n$ since we're dealing with an Erlang process. The question then is how many spanning forests rooted at j, n have a path from $i \rightsquigarrow j$? First, there is only one spanning forest for any pair of roots. Second, every such spanning forest has a path $i \rightsquigarrow j$. Consider the spanning tree $1 \rightarrow 2 \rightarrow \dots \rightarrow j \rightarrow \dots \rightarrow n$, which is just the whole graph. Every spanning 2-forest is the spanning tree with one edge removed. But removing this single edge between 1 and j would make j not be a root, because then j would have an outgoing edge to $j + 1$. So each spanning 2-forest rooted at j, n must have their missing edge between j and n , i.e., there is always a path $i \rightsquigarrow j$. It follows that $g(j, q, i) = 1$ for all j , so the denominator is $(n - 1)^m$.

$$s_m = \frac{m! \sum_{(j_1, \dots, j_m)} \left(\prod_{u=1}^m [g(j_u, q, j_{u-1})] \right)}{(n - 1)^m}$$

In the numerator, we can apply similar logic. There is always a path in the spanning 2-forest rooted at j_u and q from $j_{u-1} \rightsquigarrow j_u$ if $j_{u-1} \leq j_u$, which depends on our selection of $(j_1, \dots, j_m)$. In particular, for the product to be nonzero, each $g(j_u, q, j_{u-1})$ must satisfy this property. If they all do satisfy it, then the product equals 1. The property $j_{u-1} \leq j_u$ means the tuple $(j_1, \dots, j_m)$ is "ascending". So the value of the numerator amounts to $m!$ times the number of ascending m -tuples where the entries are chosen from $X = \{1, 2, \dots, n - 1\}$. However, we can also have repeats, for example $(j_1, \dots, j_m) = (1, 1, \dots, 1)$, and adjacent repeats in the tuple trivially satisfy the path condition.

Observe that the number of ascending j tuples is the same as the number of combinations of digits in this scheme. As an example, consider $n = 3$ and $m = 2$. Then the possible values of j are: $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$. There are 3 ascending tuples here, the only descending one is $(2, 1)$. But for any selection of digits (combination), with or without repeats, there is only one completely ascending ordering (because $\mathbb{N}$ is a total order $\odot$). So the number of combinations, in this case 3, is the same as the number of ascending tuples. Allowing repeats, the formula for choosing M things from a total of N things is given by $\binom{N+M-1}{M}$. Substituting $N = n - 1$ and $M = m$, there are

$$\binom{n+m-2}{m} = \frac{(n+m-2)!}{m!(n-2)!}$$

vertex tuples $(j_1, \dots, j_m)$ that yield nonzero products in the numerator. Each valid product equals 1, so we have

$$s_m = \frac{m! \frac{(n+m-2)!}{m!(n-2)!}}{(n-1)^m}$$

$$s_m = \frac{(n+m-2)!}{(n-2)!(n-1)^m}$$

So we know the value for any Erlang. $\square$

Sense Check: What happens for $m = 2$, the normal randomness parameter case? Formula says $s_2 = (n+2-2)! / ((n-2)!(n-1)^2) = n! / ((n-2)!(n-1)^2) = n(n-1) / (n-1)^2 = n / (n-1)$, exactly what we'd expect from Aldous-Shepp. $\odot$

Conjecture 24 (Erlang is the minimum). *The value of s_m for the Erlang graph on n vertices is a lower bound on s_m for every graph G on n vertices, in other words,*

$$s_m(E_n) \leq s_m(G)$$